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Application 3: Field-Based Bridge: Learning from In-Service Mathematics Teachers

PARTICIPANT PROFILE

Teacher J is a teacher at STEAM Academy Middle School, located at 201 Brotherton Lane, Ferguson, Missouri 63135. With nine years of teaching experience, Teacher J has developed valuable knowledge and skills in teaching mathematics to students across different grade levels, specifically from Grades 7 to 12. Throughout these years, Teacher J has gained experience in addressing diverse learning needs, explaining mathematical concepts, and helping students develop their mathematical understanding. This professional background provides valuable insights into mathematics instruction and the challenges and opportunities involved in teaching mathematics at the secondary level.

INTERVIEW CONTEXT

The interview with Teacher J took place from October 8-11, 2026, conducted as an asynchronous-online format through a Google Forms survey, where the respondent answered questions independently across the date window; its approximate total duration was 25-35 minutes, covering reading, reflecting, and responding to all interview prompts. Prior to the activity, permission was obtained from Teacher J both to participate in the interview and to take screenshots of their submitted answers, with the assurance that collected responses would be used solely for educational-related documentation and reflection.

SUMMARY OF RESPONSES

Teacher J shared valuable insights into the integration of technology in mathematics instruction. Ze explained that tools such as Desmos help students visualize geometric concepts, while platforms like DeltaMath improve efficiency in algebra practice by providing immediate feedback. These digital resources also allow students to explore and perform dynamic mathematical activities that would be difficult to accomplish using traditional pen-and-paper methods. In selecting instructional resources, Teacher J considers the lesson objectives, availability of resources, and formative assessment needs. However, ze acknowledged that some mathematical topics remain challenging to teach using the school's existing technological resources.

To address challenges related to limited access to devices, internet connectivity, and electricity, Teacher J provides printed worksheets and opportunities for students to complete their activities in class. Ze also emphasized the importance of conceptual understanding, explaining that students must not simply rely on technology to generate correct answers. Instead, ze require learners to provide step-by-step solutions and participate in peer discussions to explain their mathematical reasoning. To determine whether technology genuinely improves learning rather than merely increasing student engagement, Teacher J compares pre-assessment and post-assessment results and evaluates students' ability to solve mathematical problems independently without digital tools.

Furthermore, Teacher J developed skills in designing interactive mathematics lessons and utilizing technology-based assessment tools. Ze advised future mathematics teachers to choose accessible technological resources that align with learning objectives, prepare non-digital alternatives, and prioritize students' understanding of mathematical concepts over the completion of activities. Overall, Teacher J' s insights highlight that technology is most effective in mathematics instruction when it supports visualization, provides meaningful feedback, promotes mathematical reasoning, and remains responsive to students' diverse learning needs and circumstances.

SIGNIFICANT CLASSROOM INCIDENTS

Teacher J shared two concrete classroom experiences that demonstrate the effective integration of technology into mathematics instruction. In a Geometry lesson, Teacher J used Desmos to help students explore linear graphs and understand the relationship between equations and their graphical representations. Instead of relying solely on manually plotting points on graph paper, students adjusted the values in linear equations in real time and observed how these changes affected the slope, direction, and position of the lines. This interactive approach allowed learners to visualize mathematical relationships, identify patterns, and develop a deeper understanding of how changes in an equation influence its graph.

In an Algebra 1 lesson, Teacher J used DeltaMath to provide students with exercises on simplifying algebraic expressions and applying the properties of exponents. The platform automatically checked students' answers and provided immediate feedback when errors occurred, allowing learners to identify and correct their mistakes during practice. This helped students reinforce their understanding of algebraic procedures while enabling the teacher to monitor their progress and identify concepts that required further explanation. Additionally, the automatic checking feature reduced the time spent grading individual responses, allowing the teacher to focus more on guiding students through challenging mathematical concepts.

Overall, these experiences demonstrate how technology can enhance mathematics instruction by making learning more interactive, providing timely feedback, and supporting students' understanding of mathematical concepts. Desmos promoted visualization and conceptual understanding, while DeltaMath supported procedural fluency and independent practice. Both tools helped students engage more actively with mathematical ideas and provided opportunities for immediate exploration and correction. However, their effectiveness still depends on appropriate teacher guidance and students' ability to explain the reasoning behind their answers. When aligned with learning objectives and students' needs, technology can serve as a valuable tool for making mathematics learning more meaningful and effective.

FRAMEWORK- BASED ANALYSIS

Based on the interview responses, two appropriate frameworks were applied: the TPACK and the SAMR Model framework.

Framework 1: TPACK

The TPACK framework describes how a teacher coordinates content domain, teaching methods, and digital tools. The interview demonstrated a form of Technological Pedagogical Knowledge (TPK) and Technological Content Knowledge (TCK). The teacher chose DeltaMath because it grades math problems automatically and gives students instant feedback when they make mistakes. This saved valuable class time and allowed students to fix their errors right away.

Fourth Question Response: "DeltaMath in my Algebra 1 class to help students practice simplifying expressions and applying the properties of exponents. The technology saves time because it automatically checks students' answers and provides immediate feedback when they make mistakes."

Getting instant feedback stopped students from repeating the same mistakes over and over during their math practice. This kept the lesson focused on building real math skills rather than waiting for paper worksheets to be graded.

Framework 2: SAMR

The SAMR Model is a framework that helps teachers examine how technology changes a learning task. Based on the interview evidence, the teacher's coordinate geometry lesson utilizing Desmos achieved the level of Modification. Instead of drawing lines by hand on paper, students changed the numbers in an equation on screen to instantly see how the line shifted or changed its slope. This shifted the lesson from slow, mechanical graphing to quick, visual exploration.

Fifth Question Response: "In my Geometry class, I used Desmos to teach students how to graph lines and explore the relationships between their equations and graphs. Instead of simply plotting points on paper, students could change the values in an equation and immediately observe how the line moved or changed its slope"

This fast visual feedback helped students immediately see how changes to the equation affected the graph. It made it much easier for them to connect algebra concepts with geometric shapes.

CONSTRAINTS AND TEACHER RESPONSES

Teacher J encounters several constraints when integrating technology into mathematics instruction and applies practical strategies to address them. Regarding resource-related constraints, some mathematical topics may not have suitable technological resources available at school. To address this, Teacher J selects accessible digital tools that align with the lesson objectives rather than forcing the use of unavailable or unsuitable technologies. For access-related constraints, some students have limited access to devices, mobile data, electricity, or a stable internet connection. Teacher J responds by providing printed worksheets and allocating dedicated in-class time for digital activities, ensuring that students can complete their tasks without depending on technology at home.

In addressing instructional-related constraints, Teacher J recognizes that technology may produce correct answers without ensuring that students understand the mathematical concepts behind them. To promote deeper understanding, the teacher requires students to explain their solutions step by step and participate in partner discussions about their problem-solving strategies. For assessment-related constraints, Teacher J acknowledges that students' enjoyment and engagement in technology-based lessons do not necessarily indicate improved learning. Therefore, the teacher compares students' performance before and after instruction through quizzes, exit tickets, and independent practice, while also checking whether they can solve problems without technological assistance. Finally, to address learner-related constraints, particularly difficulties in visualizing mathematical relationships using traditional pen-and-paper methods, Teacher J uses interactive platforms such as Desmos and DeltaMath. These tools allow students to explore mathematical concepts dynamically, observe changes in representations, practice independently, and receive timely feedback, supporting their understanding of key mathematical ideas.

LESSONS FOR BEGINNING MATHEMATICS TEACHERS

The interview with Teacher J provides several valuable lessons for beginning mathematics teachers about the effective and responsible integration of technology into classroom instruction. First, teachers should select digital tools based on the lesson objectives, students' learning needs, and available resources rather than using technology simply for its novelty. They should also prepare alternative non-digital materials to ensure that learning can continue when technology is unavailable. Second, beginning teachers must consider unequal access to devices, internet connectivity, and electricity by providing printed learning materials and allocating in-class work time so that all students have opportunities to participate. Third, teachers should recognize that obtaining correct answers through digital tools does not necessarily mean that students understand the underlying mathematical concepts. Therefore, they should encourage learners to present step-by-step solutions, explain their reasoning, and engage in peer discussions to develop deeper conceptual understanding. Fourth, teachers should evaluate the actual effectiveness of technology by comparing students' performance before and after instruction through appropriate assessments rather than relying solely on their engagement or enjoyment during activities. Finally, beginning mathematics teachers should continuously develop their skills in using educational platforms such as Desmos and DeltaMath to design interactive and meaningful learning experiences. Through these practices, teachers can use technology purposefully to enhance mathematical understanding, strengthen problem-solving skills, and support diverse learners.

PROPOSED IMPROVEMENT

One feasible improvement Teacher J could adopt, while keeping the core mathematics-learning goal intact, is to add short, low-technology pre-activity warm-ups before using interactive tools such as Desmos or DeltaMath. Before students explore dynamic graphs or practice exercises digitally, they first sketch simple graphs or work through basic problems using pen-and-paper to activate prior conceptual knowledge. This practice preserves the original goal of helping learners visualize relationships and build procedural fluency; it also reduces the risk that students will rely only on the technology's output without grasping the underlying math, and offers extra support for learners with limited at-home digital access. After the warm-up, the digital tool can then extend understanding through dynamic exploration and immediate feedback, strengthening rather than replacing conceptual focus.

PERSONAL REFLECTION

Through our team's reflection on the interview with Teacher J, we gained a deeper understanding of what realistic and responsible technology integration means in mathematics instruction. As a team, we realized that effective

technology integration is not simply about using popular digital tools but about choosing appropriate resources that support learning objectives and respond to the actual conditions of the classroom. Realistic integration requires teachers to consider the availability of school technology and students' access to devices, internet connectivity, and electricity. We recognized the importance of preparing alternative learning materials, such as printed worksheets and non-digital activities, to ensure that all learners can participate in lessons even when technology is unavailable. This helped us understand that effective teaching requires flexibility, preparation, and consideration of students' different learning circumstances.

Furthermore, our team learned that responsible technology integration should prioritize students' conceptual understanding rather than simply making activities faster or more engaging. Although digital tools can provide immediate feedback and interactive learning experiences, students must still understand the mathematical processes behind their answers. Therefore, teachers should encourage learners to explain their reasoning, demonstrate their solutions step by step, and solve problems independently rather than depend entirely on technology. We also realized that teachers must evaluate the actual effectiveness of digital tools by assessing students' learning outcomes instead of relying only on their participation or enjoyment during activities. As future mathematics teachers, we recognize the importance of continuously developing our technological skills, selecting accessible and appropriate educational platforms, and preparing alternative teaching strategies to address possible classroom challenges. Overall, the interview encouraged our team to view technology as a means of improving mathematical understanding, strengthening problem-solving skills, and supporting diverse learners. We concluded that technology becomes truly meaningful when it is used purposefully, responsibly, and inclusively to enhance learning rather than simply complete classroom activities.